Purpose of the Simulator

The BESS Ramp Compliance Simulator evaluates how effectively a Battery Energy Storage System (BESS) can mitigate short-term solar power fluctuations and maintain compliance with a grid ramp-rate limit of:

±3 MW/min

The simulator uses one year of 1-minute solar generation data and models how a battery would respond to rapid changes in solar output.

The tool helps answer several important engineering questions:

- How much battery power is required to maintain ramp compliance?
- How much battery energy is required to support compliance?
- How often does the battery fail to maintain the ramp limit?
- How much energy is curtailed due to battery limitations?
- How does battery state of charge (SOC) evolve throughout the year?
- What operational trade-offs exist between power, energy, efficiency, and SOC limits?

The simulator includes both idealized battery studies and a realistic operational battery controller.

User Inputs

The simulator allows the user to define the physical characteristics of the battery system.

Input	Description
BESS Power Limit (MW)	Maximum charge or discharge power available from the battery
BESS Energy Capacity (MWh)	Total usable energy storage capacity
Initial Year SOC (%)	Battery state of charge at the start of the simulation
One-Way Efficiency	Charging and discharging efficiency applied during operation
SOC Window	Minimum and maximum allowable battery SOC

These inputs determine how aggressively the battery can respond to solar ramps and how much energy is available to support compliance.

Overall Program Flow

The simulator follows the sequence below.

Pseudocode

LOAD annual solar generation data

GENERATE annual timestamp series

CALCULATE ideal battery behavior

Assume:

Infinite power

Infinite energy

No SOC limits

No efficiency losses

CALCULATE minimum daily stored energy requirement

For each day:

Simulate ideal ramp compliance

Determine minimum energy required to avoid running empty

RUN full operational battery simulation

Apply:

Power limits

Energy limits

SOC limits

Efficiency losses

CALCULATE:

Export profile

Battery dispatch profile

SOC profile

Ramp violations

Curtailment statistics

Energy statistics

CALCULATE daily battery energy summaries

GENERATE plots and performance metrics

DISPLAY results to user

Ramp Compliance Philosophy

The simulator assumes the grid operator requires exported power to change no faster than:

±3 MW/min

Whenever solar generation changes faster than this limit:

- The battery absorbs excess power during rapid solar increases.
- The battery injects power during rapid solar decreases.

The battery acts as a buffer between solar production and the grid.

If the battery reaches a physical limitation:

- Power limit
- Energy limit
- SOC limit

then ramp compliance may no longer be achievable and a violation is recorded.

Detailed Explanation of run_sim()

The run_sim() function is the core of the simulator and represents the actual operational controller.

It evaluates battery performance minute-by-minute across the entire year.

Step 1 – Initialize the Simulation

At the beginning of the simulation:

- Battery energy is initialized using the selected starting SOC.
- SOC limits are converted into minimum and maximum allowable stored energy.
- Arrays are created to store:
    - Grid export
    - Battery dispatch
    - SOC history

Annual performance counters are also initialized.

These counters track:

- Solar generation
- Grid export
- Battery throughput
- Curtailment
- Ramp violations

Step 2 – Process Each Minute of Solar Data

The simulation then loops through every minute of the year.

For each minute:

Read current solar output

The solar value represents actual plant production before any battery action.

Step 3 – Apply the 100 MW Export Cap

Before ramp control is evaluated:

PV output is limited to 100 MW

Any solar generation above 100 MW is considered inherent curtailment.

Inherent Curtailment = PV Production - 100 MW

This curtailment occurs regardless of battery operation.

Step 4 – Calculate the Solar Ramp

The simulator compares:

‘Current capped solar output’ against ‘Previous minute export’ to determine the effective ramp rate.

raw_ramp = current_pv - previous_export

This represents the change that the grid would experience if the battery did nothing.

Step 5 – Determine Required Battery Action

The controller evaluates whether the ramp exceeds the allowable limit.

Case 1 – Ramp is within ±3 MW/min

No battery action required

Battery power remains zero.

Case 2 – Ramp exceeds +3 MW/min

Solar production is increasing too quickly.

The battery charges to absorb the excess increase.

Example:

Solar increase = +7 MW
Allowed increase = +3 MW

Battery charges at:
7 - 3 = 4 MW

This prevents the grid from seeing the full increase.

Case 3 – Ramp exceeds -3 MW/min

Solar production is decreasing too quickly.

The battery discharges to support the export level.

Example:

Solar decrease = -8 MW

Allowed decrease = -3 MW

Battery discharges:
8 - 3 = 5 MW

This reduces the export drop observed by the grid.

Step 6 – Apply Physical Battery Constraints

The battery may not always be able to provide the full charging or discharging power required to maintain perfect ramp compliance. Before any battery action is accepted, the controller evaluates four physical constraints:

- Power limit
- Available energy
- SOC operating window
- Efficiency losses

The battery's actual response is the maximum response that satisfies all of these constraints simultaneously.

6.1 Power Limit Constraint

The first limitation is the battery inverter power rating.

The controller calculates the battery power required to achieve perfect ramp compliance:

target

However, the battery cannot exceed its rated charge or discharge capability:

$-P_{max} \le P_{BESS} \le P_{max}$

where:

- P_{max} = user-selected BESS Power Limit (MW)

For example:

Required charging power = 12 MW
BESS power rating = 8 MW

The battery can only charge at:

8 MW

The remaining 4 MW cannot be absorbed and may result in additional curtailment or a ramp-rate violation.

6.2 SOC Window Constraint

The battery is not allowed to operate outside the user-defined SOC window.

The selected SOC limits are converted into allowable stored energy limits:

e_min = SOC_min × Energy Capacity
e_max = SOC_max × Energy Capacity

Examples:

SOC Window	Energy Capacity	Allowed Energy Range
10–90%	40 MWh	4 MWh to 36 MWh
20–80%	40 MWh	8 MWh to 32 MWh
30–70%	40 MWh	12 MWh to 28 MWh

The battery cannot:

- Charge above e_max
- Discharge below e_min

Even if additional ramp support is required.

6.3 Available Charging Energy Constraint

When the battery needs to charge, the controller calculates how much empty storage space remains before reaching the SOC ceiling.

Remaining energy space:

e_max − current_energy

This available energy space is converted into an equivalent maximum charging power for the current one-minute timestep:

available_charge_power = ((e_max − current_energy) × 60) / efficiency

This determines the largest charging power that can be accepted without exceeding the maximum SOC.

If the requested charging power is larger than this value, charging is reduced accordingly.

Once the battery reaches the upper SOC limit, no further charging is possible.

6.4 Available Discharge Energy Constraint

When the battery needs to discharge, the controller calculates how much usable energy remains above the SOC floor.

Available stored energy:

current_energy − e_min

This is converted into an equivalent maximum discharge power:

available_discharge_power = ((current_energy − e_min) × 60) × efficiency

This determines the largest discharge power that can be delivered without violating the minimum SOC limit.

If the requested discharge power exceeds this value, the discharge power is reduced.

Once the battery reaches the lower SOC limit, no further discharge is possible.

6.5 Efficiency Constraint

The simulator applies the user-selected one-way efficiency during all energy transfers.

Charging

When charging, not all incoming energy becomes stored energy:

Stored Energy Increase = Charge Power × Efficiency

Example:

Battery charging = 10 MW
Efficiency = 97%

For one minute:

Stored energy increase = 10 × 0.97 / 60 = 0.162 MWh

The remaining energy is lost as conversion losses.

Discharging

When discharging, additional stored energy must be removed to deliver the requested output power:

Stored Energy Decrease = Discharge Power / Efficiency

Example:

Battery discharge = 10 MW
Efficiency = 97%

For one minute:

Stored energy decrease = 10 / 0.97 / 60 = 0.172 MWh

The difference represents conversion losses.

6.6 Determining the Actual Battery Response

The requested ramp-control power is compared against all constraints simultaneously.

The actual battery response becomes:

Actual BESS Power = minimum(target, power limit, available charging/discharging power)

As a result, the battery may not always be able to provide the full ramp-support power requested by the controller.

When this occurs:

- Ramp compliance may be lost.
- Additional curtailment may occur.
- Violations may be recorded.

This behavior represents the real-world limitations of an actual battery energy storage system.

Step 7 – Update Battery Energy

After the actual battery power is determined:

Charging

Stored energy increases:

Battery Energy += Charge Power × Efficiency

Discharging

Stored energy decreases:

Battery Energy -= Discharge Power / Efficiency

The resulting energy level is converted into SOC.

Step 8 – Calculate Grid Export

The final export delivered to the grid is:

Grid Export = Capped Solar Output − Battery Charging Power + Battery Discharging Power

This is the power profile actually observed by the grid.

Step 9 – Track Curtailment

Two curtailment categories are recorded.

Inherent Curtailment

Occurs whenever:

PV > 100 MW

and is unrelated to battery limitations.

Ramp Curtailment

Occurs when:

Battery cannot absorb
all required charging power

because of:

- Power limit
- Maximum SOC
- Energy limitation

In this situation additional solar energy must be curtailed.

Step 10 – Check Ramp Compliance

The simulator evaluates whether the final export change exceeds:

±3 MW/min

If the limit is exceeded while solar production is active:

Violation Count += 1

Each violation represents one minute of non-compliance.

Step 11 – Store Results

For every minute, the simulator records:

- Export power
- Battery dispatch power
- Battery SOC

These arrays are later used to generate plots and annual statistics.

Interpretation of Results

The simulator produces two categories of outputs.

Idealized Studies

These plots assume perfect battery operation and are primarily used for sizing studies.

- Daily Ideal BESS Net Energy Required (No limitations)
- Daily Minimum Initial Stored Energy Required (Power capacity limited)

These answer:

What would be required if the battery had no operational limitations?

Operational Studies

These plots use the full run_sim() controller and represent realistic battery operation.

- Daily BESS Net Energy
- Annual Solar, Export, BESS, and SOC

These answer:

What actually happens with the battery configuration selected by the user?

The difference between the ideal and operational plots highlights the impact of real-world constraints such as finite power, finite energy, efficiency losses, and SOC limits.